
The Compression Floor: How Few-Step Decoding Collapses in Small Block-Diffusion Language Models

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Abstract

1 Block-diffusion language models accelerate generation by predicting multiple
2 tokens per denoising step, but the limits of this compression remain poorly un-
3 derstood. Therefore, we study SDAR-4B-Chat (a 4B-parameter block-diffusion
4 language model) with block size **32** on a mathematical benchmark and identify a
5 sharp **compression floor**, the limit beyond which predicting more tokens per step
6 causes generation quality to collapse, with three properties. First, generation quality
7 does not gradually decline as more tokens are predicted per step; instead, outputs
8 tend to remain stable or suddenly collapse into repetition: across pooled schedules,
9 **60%** of outputs remain largely repetition-free ($\text{repeat-4} < \mathbf{0.2}$), while **23%** collapse
10 into severe repetition loops ($\text{repeat-4} > \mathbf{0.7}$), with only **9%** in the intermediate
11 regime (repeat-4 between **0.3** and **0.6**). Second, supervised fine-tuning (SFT) can
12 worsen compression tolerance. Dense supervision on all masked positions (L_{all})
13 reduces one-token-per-step accuracy from **48%** to **8%**, whereas supervision on
14 commit-only positions (L_{commit} ; positions selected for final prediction) holds it
15 at **20%**. SFT also reduces the usable compression range from four to two tokens
16 per step. Third, on-policy self-distillation inherits this floor through trajectory
17 yield: at temperature **0.9**, the student produced **0** usable trajectories in **28** attempts,
18 while temperature **0.3** with a larger generation budget yielded **38/50**. We repli-
19 cate this yield collapse on a second model family, suggesting the phenomenon
20 is not architecture-specific. Together, these results show that few-step diffusion
21 language modeling is governed not only by the number of tokens revealed per
22 step, but also by how supervision and sampling interact with the model’s compres-
23 sion floor. Our findings highlight supervision density and sampling strategy as
24 two practical levers for making accelerated diffusion-LM training reliable. The
25 code is available at [https://anonymous.4open.science/r/Compression_](https://anonymous.4open.science/r/Compression_Floor--Diffusion_Language_Model-1BC2/README.md)
26 [Floor--Diffusion_Language_Model-1BC2/README.md](https://anonymous.4open.science/r/Compression_Floor--Diffusion_Language_Model-1BC2/README.md).

27 1 Introduction

28 Diffusion language models generate text through iterative denoising, progressively refining a partially
29 specified sequence while predicting multiple tokens in parallel at each step. Block-diffusion variants
30 such as SDAR [Cheng et al., 2025] and TraDo [Wang et al., 2025] extend this idea by decoding tokens
31 in fixed-size blocks in parallel while remaining autoregressive across blocks. This offers a simple
32 route to faster generation: increasing the number of tokens decoded per step reduces the number of
33 forward passes required. *Understanding how far this compression can be pushed is therefore critical*
34 *for determining the practical speedup and compute efficiency of diffusion language models.*

35 However, this speedup raises a fundamental question: *how far can we increase the number of tokens*
36 *decoded per step before generation quality breaks down?* We study this question using SDAR-4B-
37 Chat with block size 32 on mathematical reasoning problems from DeepMath-103K [He et al., 2025].

Recent work has pursued *few-step diffusion language modeling* along two complementary directions. Both masked and block-diffusion hybrids such as SDAR [Cheng et al., 2025] and TraDo [Wang et al., 2025] reduce sequential depth by revealing multiple tokens in parallel, while training-free methods such as Fast-dLLM [Wu et al., 2025] further accelerate inference without modifying the model. These approaches expose a fundamental trade-off between *generation speed and the number of denoising steps*, and recent analyses attribute few-step degradation to errors introduced when multiple tokens must be predicted simultaneously.

A second line of work attempts to recover this lost quality through *distillation*. T3D [Zhang et al., 2026] distills on rollout trajectories produced under the target decoding procedure, while OPSD [Zhao et al., 2026] introduces *on-policy self-distillation*, in which teacher and student share weights and the teacher conditions on privileged information while both are evaluated on the student’s own rollout. However, these approaches implicitly require the student to produce sufficiently useful trajectories for training.

In contrast, we focus on the regime where this prerequisite itself breaks down: we characterize the *compression floor* at the level of individual generations and study how supervised fine-tuning and sampling choices move or expose this floor. Thus, rather than proposing another acceleration or distillation method, our work provides an empirical characterization of when few-step diffusion generation remains trainable and when it collapses.

2 Experimental Setup

Model and data: We conduct our experiments on SDAR-4B-Chat [Cheng et al., 2025], a 4B-parameter block-diffusion language model with block size 32 (b32) and block size 16 (b16), obtained by adapting an autoregressive base model. Training and evaluation instances are drawn from DeepMath-103K [He et al., 2025], which provides verified final answers together with three R1-generated solution traces per problem. We use a fixed held-out benchmark of 40 problems, stratified by dataset difficulty into 20 easy, 10 medium, and 10 hard instances. This subset was selected to provide balanced coverage across difficulty levels while keeping the extensive multi-schedule decoding and training experiments computationally feasible. Unless otherwise specified, fine-tuning is performed with LoRA adapters of rank 32 while keeping the base model frozen.

Decoding protocol: For a block containing 32 masked positions, we decode t tokens per denoising step by committing the t highest-confidence masked positions. Thus, $t = 1$ requires 32 denoising steps per block, whereas $t = 32$ completes the block in a single step. We evaluate schedules spanning $t \in \{1, 2, 4, 8, 16\}$ and use greedy decoding unless otherwise stated. Generations are limited to 1,536 tokens.

Evaluation metrics: We measure both task completion and generation stability. *Boxed rate* is the fraction of generations containing a final boxed answer, while *correct rate* additionally requires the extracted boxed answer to match the verified reference answer. To quantify degenerative repetition, we use *Repeat-4*, defined as one minus the fraction of unique 4-grams in a generation. Thus, lower values indicate greater diversity, while values approaching one correspond to severe repetition.

Evaluation protocols: We use two evaluation harnesses for distinct experimental comparisons while keeping the underlying benchmark and prompts fixed. The decoding-grid experiments in Section 3.1 and the SFT control comparison in Section 3.2 use a common grid-based evaluator with an explicit repetition-collapse stopping criterion. The three-stage curriculum comparison in Section 3.2 uses the training harness associated with the corresponding fine-tuning runs and reports a more permissive string-match accuracy. Because the stopping and matching criteria differ, results from the two protocols are treated as separate measurements and are never combined within a single comparison. Where needed, the base model is evaluated independently under each protocol to provide the appropriate reference point.

3 Results

3.1 The floor is a per-sample phase transition

Table 1 reports the base model at block 32 across schedules. Accuracy holds up to $t = 2$, degrades at $t = 4$, and reaches zero between $t = 4$ and $t = 8$. The collapse-detector stop fraction mirrors this

119 cliff, rising from 0 at $t \leq 2$ to 0.62 at $t = 16$. Block size 16 shows the identical cliff location (correct
120 0.20 at $t = 4$, 0.00 at $t = 8$).

Table 1: Base SDAR-4B-Chat at block sizes 16 and 32 across decoding schedules (40 problems per cell, greedy). The accuracy cliff sits between 4 and 8 tokens per step. Collapse is the fraction of generations halted by the repetition detector.

Block	Metric	$t=1$	$t=2$	$t=4$	$t=8$	$t=16$
16	correct	.425	.375	.200	.000	.000
	repeat-4	.152	.181	.403	.579	.684
	collapse	.000	.000	.100	.500	.500
32	correct	.375	.250	.150	.000	.000
	repeat-4	.161	.156	.268	.386	.590
	collapse	.000	.000	.050	.275	.625

121 The mean curves hide the structure. Figure 2 pools per-problem repeat-4 over all schedules. The
122 distribution is bimodal. At block 32, 60% of outputs sit below 0.2, 23% sit above 0.7, and only 9%
123 occupy the middle band from 0.3 to 0.6. Individual generations do not degrade gradually as the
124 schedule coarsens. They either hold together or fall into a loop, and coarser schedules only shift mass
125 between the two modes. This matches the factorization-error account of few-step decoding [Zhang
126 et al., 2026]: once simultaneously committed tokens must be predicted independently, an unlucky
127 commitment locks the block, and everything after it, into repetition. The cost is also difficulty-
128 selective. At block 32 and $t = 1$, medium-tier boxed rate is 0.60, and it is the first tier to reach zero
129 as t grows.

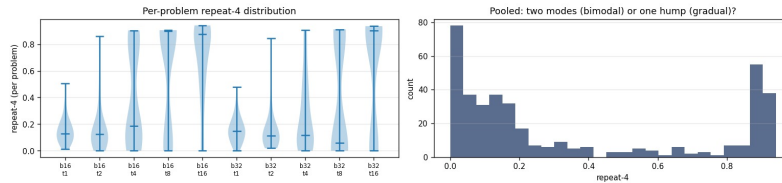


Figure 2: Per-problem repeat-4 for the base model, pooled over schedules. The distribution has a fluent mode and a collapsed mode with little mass between them. Cell means rise with tokens per step only because the collapsed fraction grows.

130 3.2 SFT narrows the window, and supervision density and dataset size decide how much

131 We fine-tuned the model three ways on DeepMath solutions, always with LoRA, and benchmarked
132 every checkpoint on the fixed 40 problems.

133 **SFT shifts the cliff inward.** A three-stage SFT run that supervised all currently masked positions
134 per denoising state was re-benchmarked across the full schedule grid (Figure 3). Relative to the base
135 model at block 32, its ceiling fell (correct 0.375 to 0.175 at $t = 1$) and its cliff moved one octave
136 earlier, from between $t = 4$ and $t = 8$ to between $t = 2$ and $t = 4$. Redundancy rose at every
137 schedule. Reference-solution coverage at $t = 1$ rose above base (0.506 versus 0.446) while accuracy
138 fell, so the model learned to imitate reference phrasing at the cost of finishing problems. Medium-tier
139 accuracy was zero at every schedule, including $t = 1$.

140 **A three-stage curriculum lets us ask whether this collapse is the objective, or the data.** We ran
141 two matched three-stage curricula (stages at 4, 2, and 1 tokens per step, 500 problems and 2 epochs
142 per stage) that differ in both the loss and the data schedule. The dense arm supervises every currently
143 masked position at each denoising state, in contrast to the standard masked-diffusion objective [Sahoo
144 et al., 2024], and reuses the same 500-problem pool across all three stages, giving each problem 6
145 effective epochs of exposure. The commit-only arm supervises only the t highest-confidence masked
146 positions, that is, exactly those the decoder commits at that step and draws a fresh 500-problem pool
147 for each stage (1,500 problems total, no reuse), specifically to remove the overfitting we observed
148 in the dense arm. Because the two arms differ in loss and data simultaneously, this is not a clean

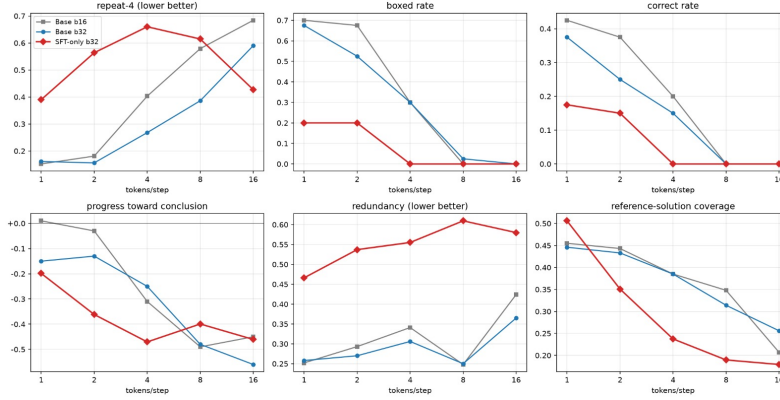


Figure 3: Base model versus the dense-supervision SFT control at block 32, across common schedules. SFT lowers the accuracy ceiling, moves the collapse cliff one octave earlier, and raises redundancy everywhere, while increasing surface similarity to reference solutions at $t = 1$.

149 ablation of the objective; it directly tests whether fixing the data-reuse problem is enough to recover
 150 base performance.

151 It is not. Table 2 shows that dense supervision degrades monotonically and severely as the curriculum
 152 progresses (48% to 8% correct at $t = 1$), consistent with the repeated pool driving memorization-then-
 153 collapse rather than a property of dense supervision per se. Removing the reuse in the commit-only
 154 arm recovers 10 to 18 points at every stage and roughly halves the repetition increase, confirming that
 155 data reuse explains a large part of the dense arm’s failure. But the commit-only arm’s best checkpoint
 156 still trails the base model by 18 points at $t = 1$ (30% versus 48%), despite using fresh, non-repeated
 157 data at every stage. So while overfitting on a reused pool is the dominant driver of the dense arm’s
 158 collapse, it does not account for all of the degradation: fine-tuning on this curriculum degrades the
 159 model even when overfitting is controlled for, and we cannot attribute that residual gap to the loss
 160 function alone given the remaining confounds below.

161 One might object that the dense arm’s disadvantage is simply an artifact of our attribution, not a
 162 property of the objective: perhaps dense supervision would match or exceed commit-only if it too
 163 were given 1,500 unique problems instead of 500 reused ones. Stage 1 speaks to this. At stage 1,
 164 neither arm has yet reused any data: the dense arm has seen its 500-problem pool for the first time,
 165 and the commit-only arm has seen its first fresh 500-problem pool, so the two checkpoints are trained
 166 on equally novel, equally sized data at this point in the curriculum. Dense is nonetheless already well
 167 behind commit-only (20% versus 30% correct at $t = 1$), and the gap widens at every subsequent
 168 stage as the dense arm’s pool reuse compounds. That the disadvantage is present before any reuse
 169 has occurred, and grows monotonically once it does, is consistent with dense supervision being a
 170 poor objective in its own right rather than merely a data-starved one; more data would need to reverse
 171 an already-negative and worsening trend, not just extend a currently-even one.

172 Two further observations hold regardless of this attribution question. First, the medium tier collapses
 173 to zero under nearly every fine-tuned checkpoint in both arms, so difficulty-selective fragility appears
 174 in both the overfit and non-overfit conditions. Second, the commit-only arm shows a schedule
 175 crossover absent under dense supervision: its final stage is the best checkpoint at $t = 2$ (30%) and
 176 the worst at $t = 1$ (20%), so the residual failure changes shape, not just size.

177 3.3 Self-distillation inherits the floor through trajectory yield

178 On-policy self-distillation requires the student model to generate useful trajectories before training
 179 can begin. For our math tasks, we consider a rollout useful if it reaches a boxed final answer within
 180 the generation budget. Table 3 shows that this yield depends strongly on the sampling settings.
 181 With temperature 0.9 and a 768-token budget, we obtained 0 usable rollouts from 28 attempts with
 182 SDAR-4B-Chat. When we lowered the temperature to 0.3 and increased the budget to 1,536 tokens,
 183 we obtained 38 usable rollouts from 50 attempts, including 13 correct ones. The failed rollouts
 184 were not necessarily poor generations: their mean repeat-4 score was only 0.17, and none produced

Table 2: Supervision-density comparison on the fixed benchmark (training-notebook evaluator; the base row uses the same evaluator). Accuracy is the lenient match rate; repeat-4 is reported at $t=1$. The commit-only objective recovers 10 to 18 points at every stage but does not restore base accuracy.

Checkpoint	Dense (all masked)		commit-only (top- t)		Δ acc.
	$t=1$	repeat-4	$t=1$	repeat-4	
base	48%	.185	48%	.185	–
stage 1 ($t=4$)	20%	.436	30%	.274	+10
stage 2 ($t=2$)	10%	.608	28%	.247	+18
stage 3 ($t=1$)	8%	.497	20%	.350	+12

an incorrect boxed answer. They simply failed to reach a conclusion within the allowed budget. Because temperature and budget were changed together, we cannot separate their contributions. The failure signature, low repetition and no incorrect boxed answers, points to the budget as the binding constraint. We observed a similar issue with another model, TraDo-4B [Wang et al., 2025], where temperature 0.9 produced no boxed rollouts in the first 36 attempts. These results show that trajectory yield is strongly affected by the sampling settings and should be measured before running self-distillation.

This issue is also important for existing diffusion OPSD methods. [Luo et al., 2026] build the self-teacher from the student’s own on-policy trajectories, which are the same type of trajectories that we screen here. On GSM8K-scale tasks, they report that using a pass@ k budget of $k=8$ works better than $k=1$, with $k=1$ showing a measurable performance drop (Table 8 of their work). Our results provide a possible explanation for why a larger sampling budget is useful: usable trajectories can be rare under some sampling settings. In our SDAR-4B-Chat-b32 experiment, temperature 0.9 produced zero usable trajectories with a 768-token budget. In such a case, simply increasing the number of samples does not solve the problem unless the generation settings also allow the model to reach a conclusion. The required sampling budget therefore depends on the model and the sampling regime. We argue that trajectory yield should be measured and reported before applying self-distillation, because a distillation method that requires usable student trajectories cannot work when those trajectories are not produced in the first place.

We also attempted distillation on the 38 accepted trajectories, with a teacher conditioned on a reference solution and given a finer schedule than the student. The loss collapsed to near zero within a few optimizer steps. Inspecting per-position distributions showed why: the privileged conditioning moved the teacher’s distribution only marginally away from the student’s, so the clipped KL target carried almost no gradient. We report this as a negative result about this particular teacher construction at 4B scale, not about on-policy self-distillation in general; a teacher whose advantage comes from within-block lookahead rather than external context remains untested here.

Table 3: Trajectory yield of screening rollouts under different sampling parameters. Usable means the rollout contains a boxed final answer. Yield, the precondition for on-policy self-distillation, is set by sampling parameters rather than by model capability.

Model	Temp. / budget	Usable	Correct	Mean repeat-4
SDAR-4B-Chat-b32	0.9 / 768	0 / 28	0 / 28	0.17
SDAR-4B-Chat-b32	0.3 / 1,536	38 / 50	13 / 50	–
TraDo-4B	0.9 / 768	0 / 36	0 / 36	–

4 Limitations and conclusion

Our results have several limitations. We use a single 40-problem benchmark and one random seed, giving accuracy estimates with binomial uncertainty of about ± 15 points; the results should therefore be viewed as evidence of large effects rather than precise estimates. All fine-tuning experiments use LoRA on a single 4B model, so behavior may differ with full fine-tuning or larger models. The dense and commit-only training arms also differ in data reuse and target cleanliness, preventing us from attributing their performance gap solely to the training objective. In particular, most reference solutions exceed the training length limit, producing truncated targets without stop tokens and

potentially contributing to the non-termination observed across fine-tuned checkpoints. Finally, we do not measure the final performance after self-distillation: although 38 trajectories were accepted, privileged-teacher distillation rapidly drove the loss near zero, suggesting that the teacher provided little additional information beyond the student’s distribution. Our results therefore establish the conditions needed for distillation, but not its eventual benefit.

Overall, we find that the compression limit of a small block-diffusion language model is not a smooth decline in performance. Instead, individual samples undergo a sharp transition from successful generation to repetitive collapse. Fine-tuning worsens this limit across all variants tested, with supervision density playing an important role: commit-only supervision reduces the degradation relative to dense supervision but does not eliminate it. We further find that self-distillation depends critically on obtaining a sufficient yield of usable trajectories, which is strongly affected by sampling temperature and generation budget. Future work should therefore report per-sample collapse rates, supervision density, and trajectory yield alongside final distillation performance.

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A Additional Related Work

Block-diffusion language models and few-step decoding. Masked diffusion language models [Sahoo et al., 2024, Nie et al., 2025] and block-diffusion hybrids [Arriola et al., 2025, Cheng et al., 2025, Wang et al., 2025] trade sequential depth for parallel width. Training-free accelerators such as Fast-dLLM [Wu et al., 2025] push the same trade-off at inference time. T3D [Zhang et al., 2026] attributes few-step degradation to factorization error in the token-independent reverse process, which grows as steps shrink. Our work differs in scope: we do not propose an accelerator. We characterize the shape of the failure at small scale, per sample rather than in the mean, and quantify how post-training moves it.

Self-distillation for fewer steps. SDTT [Deschenaux and Gülçehre, 2025] distills a many-step teacher into a few-step student. T3D [Zhang et al., 2026] distills on rollout trajectories produced under the target decoding procedure. OPSD [Zhao et al., 2026] makes distillation on-policy for autoregressive models: teacher and student share weights, the teacher conditions on privileged information, and both are evaluated on the student’s own rollout. A concurrent work adapts on-policy self-distillation to diffusion models on LLaDA-8B [Luo et al., 2026]. We differ from all of these in what we measure. Prior work reports post-distillation accuracy on students that produce usable trajectories. We measure the precondition itself: whether a small student near the floor produces trainable trajectories at all, and what controls that.

B Formal Metric Definitions

Boxed rate and correct rate. For a generation g and reference answer a^* , let $\text{box}(g)$ denote the extracted contents of the final `\boxed{\}` expression in g , or $\perp$ if none is present. Boxed rate over a set of N problems is $\text{Boxed} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{box}(g_i) \neq \perp]$. Correct rate additionally requires an exact match to the verified reference, $\text{Correct} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{box}(g_i) = a_i^*]$. Correct rate is a subset of boxed rate by construction.

Repeat-4. For a generation g tokenized into n -grams of length 4, let $U_4(g)$ be the number of distinct 4-grams and $C_4(g)$ the total count of 4-grams. Repeat-4 is $\mathcal{R}_4(g) = 1 - U_4(g)/C_4(g)$. Lower values indicate greater lexical diversity; values approaching 1 indicate that the generation is dominated by a small set of repeated 4-grams.

Collapse flag. A generation is flagged as *collapsed* if the online repetition detector halts decoding before the token budget or EOS is reached. The collapse rate over N problems is the fraction so flagged, $\text{Collapse} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\text{halted by repetition detector}]$.

Reference-solution coverage. Let $S(g)$ and $S(a_{\text{trace}}^*)$ be the sets of unique content n -grams (stopwords removed) in the generation and in the reference reasoning trace, respectively. Coverage is the fraction of reference n -grams reproduced in the generation, $\text{Coverage}(g) = |S(g) \cap S(a_{\text{trace}}^*)| / |S(a_{\text{trace}}^*)|$. Coverage is reported as the mean over the benchmark. Higher coverage indicates closer surface alignment with the reference solution and is measured independently of correctness.

Redundancy. Redundancy quantifies within-generation repetition at the sentence or reasoning-step level, distinct from the token-level Repeat-4 metric. Let $\{s_1, \dots, s_k\}$ be the segmented reasoning steps of a generation. Redundancy is the mean pairwise similarity across steps, $\text{Redundancy}(g) = \binom{k}{2}^{-1} \sum_{j < l} \text{sim}(s_j, s_l)$, where $\text{sim}(\cdot, \cdot)$ is a lexical overlap similarity (unique-token Jaccard) between step pairs. Higher values indicate the generation is looping over near-duplicate reasoning steps rather than advancing.

Progress toward conclusion. This metric scores whether successive reasoning steps move monotonically toward a final boxed answer, as opposed to stalling or reversing. For a generation segmented into steps $s_1, \dots, s_k$, let $d(s_j)$ be a step-wise progress signal (e.g., proximity of the step’s stated intermediate quantities to the final boxed value, or increasing coverage of the reference solution up to step j). Progress is the mean step-to-step change, $\text{Progress}(g) = \frac{1}{k-1} \sum_{j=1}^{k-1} [d(s_{j+1}) - d(s_j)]$. Positive values indicate the generation is converging toward a conclusion; negative values indicate

318 stalling, backtracking, or drift, which is common in collapsed generations that never reach a boxed
319 answer.

320 **Trajectory yield.** For on-policy self-distillation screening (Section 3.3), a rollout is *usable* if
321 $\text{box}(g) \neq \perp$ within the sampling budget. Yield at a given temperature τ and budget B is
322 $\text{Yield}(\tau, B) = \frac{1}{M} \sum_{i=1}^M \mathbb{I}[\text{box}(g_i) \neq \perp]$ over M screening attempts. Yield is a precondition
323 for downstream distillation and is measured independently of correctness.

324 **Dense loss and commit loss.** For a block containing L masked-position candidates at a given
325 denoising state, let $M_t \subseteq \{1, \dots, L\}$ denote the currently masked positions at step t , and let
326 $K_t \subseteq M_t$ denote the top- t highest-confidence positions among M_t , i.e., exactly those positions
327 committed at that step. The two supervision objectives used in Section 3.2 differ only in which
328 positions they sum the per-position loss $\ell(\cdot)$ over. The dense (all-position) loss supervises every
329 currently masked position in the block,

$$L_{\text{all}} = \frac{1}{|M_t|} \sum_{i \in M_t} \ell(y^i, \hat{g}^i),$$

330 while the commit (commit-only) loss restricts supervision to the positions actually committed at that
331 step,

$$L_{\text{commit}} = \frac{1}{|K_t|} \sum_{i \in K_t} \ell(y^i, \hat{g}^i).$$

332 Because $K_t \subseteq M_t$, L_{commit} is computed over a subset of the positions used by L_{all} ; the dense objective
333 supervises positions that will remain masked and be revised at later steps, while the commit objective
334 supervises only the positions that are finalized at the current step.

335 C Hyperparameters and Training Details

336 C.1 SFT Training

337 We use the same base model and LoRA configuration for both the dense (supervise-all) and commit-
338 only SFT arms. Unless otherwise stated, all hyperparameters are kept identical between the two arms.
339 The main difference is the supervision objective: the dense arm supervises every currently masked
340 position (L_{all}), while the commit-only arm supervises only the top- t positions committed at that step
341 (L_{commit}).

Table 4: Hyperparameters used for the dense and commit-only SFT arms.

Hyperparameter	Dense (supervise-all)	commit-only
Base model	SDAR-4B-Chat-b32	SDAR-4B-Chat-b32
Block size	32	32
LoRA rank	32	32
LoRA α	64	64
LoRA dropout	0.05	0.05
LoRA target modules	q/k/v/o_proj, gate/up/down_proj	q/k/v/o_proj, gate/up/down_proj
Optimizer	AdamW	AdamW
Learning rate	1×10^{-5}	1×10^{-5}
LR schedule	cosine + warmup	cosine + warmup
Warmup steps	10	10
Epochs per stage	2	2
Number of stages	3	3
Batch size	1	1
Gradient accumulation	4	4
Effective batch size	4	4
Maximum target tokens	384	384
GPU	H-100 (MiG slice, 12 GB)	Nvidia RTX Pro (96 GB)

342 The three SFT stages use the same overall training setup, with the curriculum progressing through
343 $t = 4$, $t = 2$, and $t = 1$ tokens per step. Each stage is trained for two epochs. The effective batch
344 size is four sequences, obtained from a per-step batch size of one and gradient accumulation over
345 four steps.

The dense and commit-only arms were implemented with the same optimizer, learning rate, scheduler, LoRA configuration, batch mechanics, and target length. The commit-only arm differs in which positions contribute to the loss. This controlled setup allows us to compare the effect of supervision density, while the differences in data sourcing and target preprocessing described in Section 3.2 remain important confounds.

C.2 Compute and Runtime

The SFT-only control experiment was run on a single NVIDIA H100 GPU with a 12 GB MIG slice for approximately 40 hours. The remaining experiments, including the other SFT and screening runs, were performed on a single Google Colab RTX Pro GPU for approximately 23 hours. No distributed data parallelism or model sharding was used in these experiments.

Table 5: Compute resources used for the experiments. Runtime values are approximate wall-clock runtimes for the corresponding experimental runs.

Experiment	Hardware	Runtime
SFT-only control	1 × H100, 12 GB MIG slice	~40 hours
Other experiments	1 × Nvidia RTX Pro 6000 GPU	~23 hours

All experiments use a single GPU. The GPU model for the SFT-only control was an H100 12 GB MIG slice, while the other experiments were run on a Nvidia RTX Pro 6000 GPU. Runtime differences therefore reflect both the computational workload and the available hardware.

D Assets and Licenses

Assets and licenses. Table 6 lists every external asset we use. All are publicly available, and our use of each falls within its stated terms.

Table 6: External assets used in this work.

Asset	Role	License
SDAR-4B-Chat [Cheng et al., 2025]	Primary model under study	Apache 2.0
TraDo-4B [Wang et al., 2025]	Replication model (Section 3.3)	MIT
DeepMath-103K [He et al., 2025]	Training and evaluation data	MIT

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